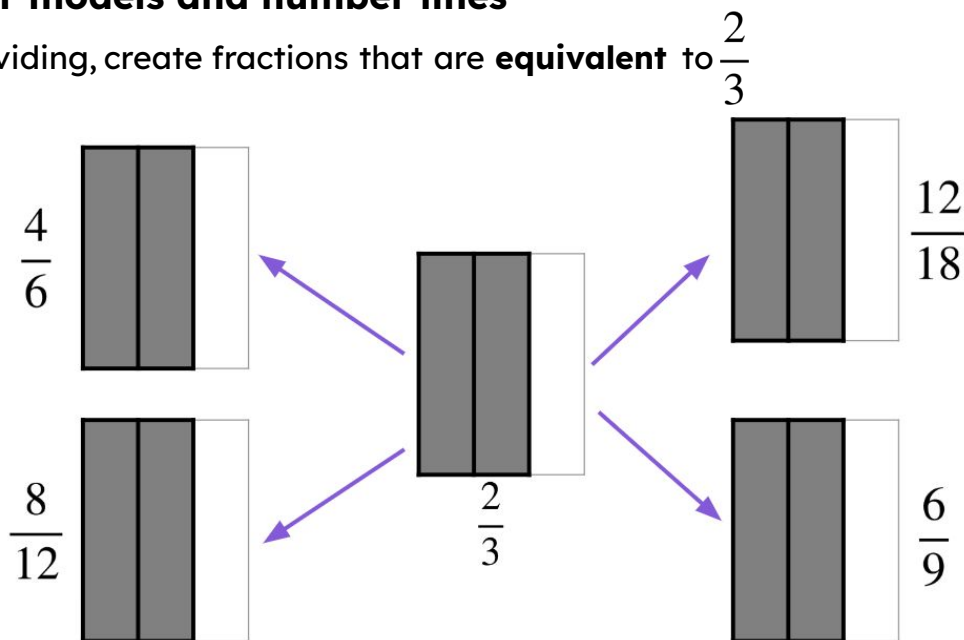




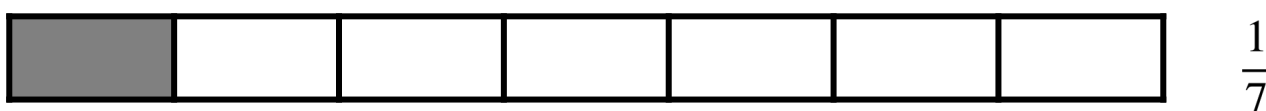
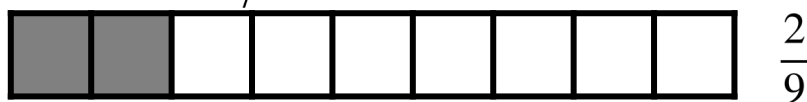
The identity property of multiplication and division

Task A: Bar models and number lines

1) By subdividing, create fractions that are **equivalent** to $\frac{2}{3}$



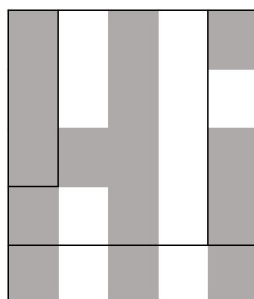
2) Sam drew a bar to show $\frac{2}{9}$
 Sam then drew another bar underneath
 which shows $\frac{1}{7}$



Sam then said, "It shows is **equivalent** $\frac{2}{9}$ to $\frac{1}{7}$ "
 Is Sam correct? Explain why.

3) Andeep says $\frac{1}{2}$ is shaded. Is he correct?

You can show your working out on the diagram.



Task B: Property of multiplication

1) Fill in the gap to show the fraction, working out and **equivalent fraction**.

a) $\frac{4}{9} = \frac{\boxed{}}{\boxed{}} = \frac{8}{18}$

c) $\frac{\boxed{}}{3} = \frac{1 \times \boxed{}}{3 \times \boxed{}} = \frac{8}{24}$

b) $\frac{7}{10} = \frac{\boxed{} \times 4}{\boxed{} \times 4} = \frac{\boxed{}}{\boxed{}}$

d) $\frac{12}{\boxed{}} = \frac{12 \times \boxed{}}{\boxed{} \times 3} = \frac{\boxed{}}{33}$

2) Shade in the equivalent fraction to $\frac{1}{3}$ to reveal a picture.

$\frac{2}{6}$	$\frac{1}{3}$	$\frac{4}{12}$	$\frac{3}{9}$	$\frac{4}{12}$	$\frac{3}{9}$	$\frac{30}{90}$	$\frac{10}{30}$
$\frac{1}{3}$	$\frac{4}{7}$	$\frac{3}{5}$	$\frac{8}{11}$	$\frac{2}{4}$	$\frac{1}{2}$	$\frac{3}{5}$	$\frac{5}{15}$
$\frac{10}{30}$	$\frac{8}{11}$	$\frac{30}{90}$	$\frac{2}{4}$	$\frac{4}{7}$	$\frac{30}{90}$	$\frac{4}{7}$	$\frac{30}{90}$
$\frac{5}{15}$	$\frac{3}{5}$	$\frac{2}{4}$	$\frac{2}{4}$	$\frac{3}{5}$	$\frac{1}{2}$	$\frac{8}{11}$	$\frac{2}{6}$
$\frac{30}{90}$	$\frac{1}{2}$	$\frac{30}{90}$	$\frac{4}{7}$	$\frac{8}{11}$	$\frac{30}{90}$	$\frac{1}{2}$	$\frac{10}{30}$
$\frac{2}{6}$	$\frac{1}{2}$	$\frac{3}{9}$	$\frac{4}{12}$	$\frac{3}{9}$	$\frac{1}{3}$	$\frac{80}{110}$	$\frac{5}{15}$
$\frac{5}{15}$	$\frac{8}{11}$	$\frac{8}{11}$	$\frac{8}{11}$	$\frac{3}{5}$	$\frac{8}{11}$	$\frac{2}{4}$	$\frac{2}{6}$
$\frac{30}{90}$	$\frac{10}{30}$	$\frac{1}{3}$	$\frac{4}{12}$	$\frac{3}{9}$	$\frac{4}{12}$	$\frac{3}{9}$	$\frac{1}{3}$

3) Here is a 5 by 5 multiplication grid.

1	2	3	4	5
2	4	6	8	10
3	6	9	12	15
4	8	12	16	20
5	10	15	20	25

Jacob uses it to help write the **equivalent fractions** to $\frac{2}{5}$

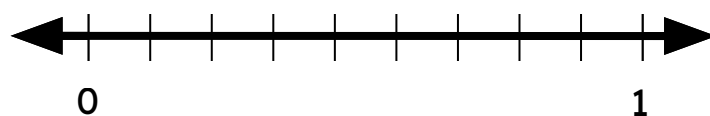
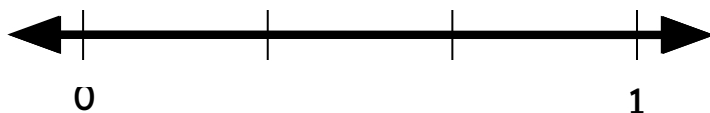
He sees

$$\frac{2}{5} = \frac{4}{10} = \frac{6}{15}$$

a) Using the grid, can you find two more **equivalent fractions** to $\frac{2}{5}$?

b) Using the grid, find **equivalent fractions** to $\frac{1}{4}$

4) Show $\frac{2}{3} = \frac{6}{9}$ using numbers lines, bar models and multiplication of 1



$$\frac{2}{3} = \frac{\boxed{}}{\boxed{}} = \frac{6}{9}$$

